

Open-Eye EEG for Biometric Security

A Deep Learning Approach



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Abstract

What is EEG Biometric Authentication?

- Utilizes unique brain activity patterns for identification.
- Non-visible and user-specific
- Nearly impossible to forge compared to fingerprints or facial recognition.

Key Methods:

- Data collected using an EEG headset under Open-Eye protocols
- Processed EEG segments analysed using a Convolutional Neural Network (CNN).
- Each trial segmented into 5-second windows for subject identification

Focus Areas:

- EEG preprocessing pipeline: filtering, normalisation, downsampling, and segmentation
- CNN architecture optimised for **spatial and temporal features**
- Use of **majority voting** to improve subject-level predictions.

Why EEG for Biometric Security?

The Problem with Conventional Biometrics:

- Fingerprints and facial recognition can be copied or spoofed.
- Physical traits are visible, permanent, and exposed to risk.
- Breaches can be irreversible (you can't change your fingerprint, “technically”).

Why EEG Signals?:

- Captures **invisible** brainwave patterns (unique to each person).
- Constantly changing and **internal**, making it harder to fake or steal.
- Even short brain activity windows can reflect deep personal traits.

Why Focus on Open-Eye EEG?

- More practical than eyes-closed or stimulus-based tasks.
- Reflects real-world, alert-state brain activity — closer to real usage.

Research Question & Hypothesis

Research Question:

Can a deep learning model, specifically a Convolutional Neural Network (CNN), accurately identify individuals based on short segments of EEG data collected under open-eye protocols?

Hypothesis:

A CNN trained on 5-second open-eye EEG segments will learn spatial and temporal brainwave patterns unique to individuals, enabling reliable subject-level biometric authentication.

Literature Survey



What Studies Show:

Importance of preprocessing (e.g., Z-score normalization).



Examples of Prior Research:

Test-retest reliability in resting EEG (88% sensitivity, 99.5% specificity).

Use of movement-related EEG for biometrics (up to 98% single session accuracy).

Literature Survey

Paper Reference	Users	Channels	Protocol	Type of Stimuli	Features	Classifier	Performance	Sessions
[1]	55	60	Resting state with eyes closed	N/A	Z-scores of spectral parameters	Generalized Linear Model (GLM)	Sensitivity: 88%, Specificity: 99.5%	2 sessions, 12–40 months apart
[2]	9	53	Resting state and movement-related EEG	Voluntary movements	Frequency-Zooming Auto-Regression (FZ-AR)	Mahalanobis distance classifier	Up to 98% (single session), 87.1% (cross-session)	2 sessions, ~1 year apart
[3]	45	56	Resting state with eyes closed	N/A	AR coefficients and reflection coefficients	Polynomial regression	Up to 98.73%	1 session
[4]	21	14	Resting state with eyes closed	N/A	Raw EEG data	ResNet, Inception, EEGNet	EEGNet: 86.74%, Inception: 70.18%, ResNet: 63.21%	3 sessions
[5]	10	64	Resting state with eyes open and closed	N/A	Raw EEG data	CNN	REO: 88%, REC: 86%, REO+REC: 82%	1 session
[6]	157	28	Resting state	N/A	Raw EEG data	GSLT-CNN	96% accuracy	Multiple sessions combined
[7]	21	8	Resting state with escalating cognitive load (serious game)	Resting and game-based	Raw EEG data	1D-CNN (Variant B: 4 convolutional layers)	Rest: 99.46%, Task Fusion: >93%	Multiple sessions combined
[8]	109	64	Resting state with eyes open and closed	N/A	Raw EEG data segmented into matrices	Deep CNN with 11 layers	98.54% accuracy	Single session
[9]	108	56	Resting state with eyes open and closed	N/A	PSD and Spectral Coherence (COH)	Mahalanobis distance-based classifier	EC: 100%, EO: 100%	Single session

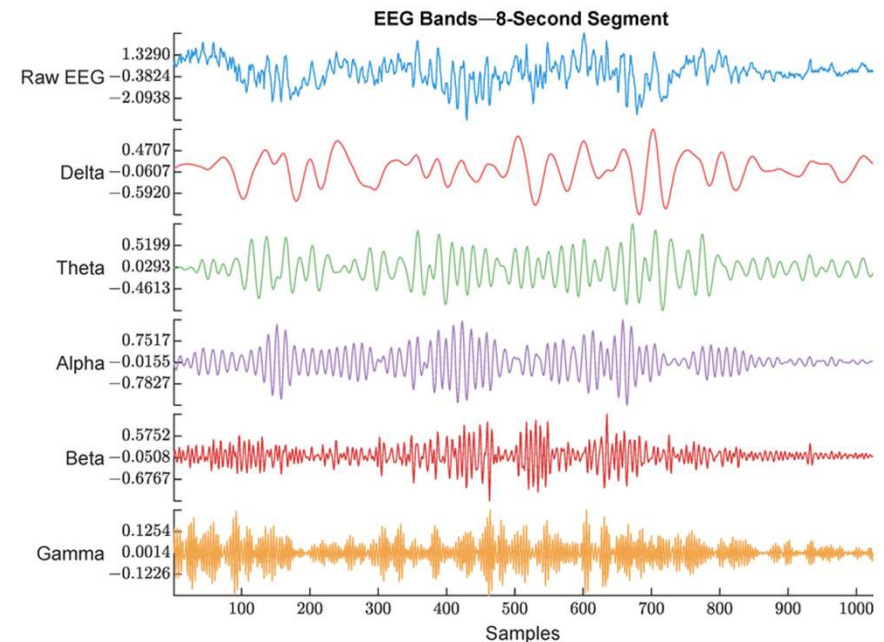
What is EEG?

What is EEG?

- Records electrical activity of the brain
- Captured using electrodes on the scalp
- Measures signals in real-time from different brain regions
- Safe, non-invasive, and cost-effective
- Used in medicine, neuroscience, and now biometrics

What Does EEG Measure?

- Brainwaves: rhythmic patterns of electrical activity
- Frequency bands:
 - **Delta (0.5–4 Hz)** – deep sleep
 - **Theta (4–8 Hz)** – drowsiness
 - **Alpha (8–13 Hz)** – relaxed, awake
 - **Beta (13–30 Hz)** – active thinking
 - **Gamma (30+ Hz)** – high-level cognition

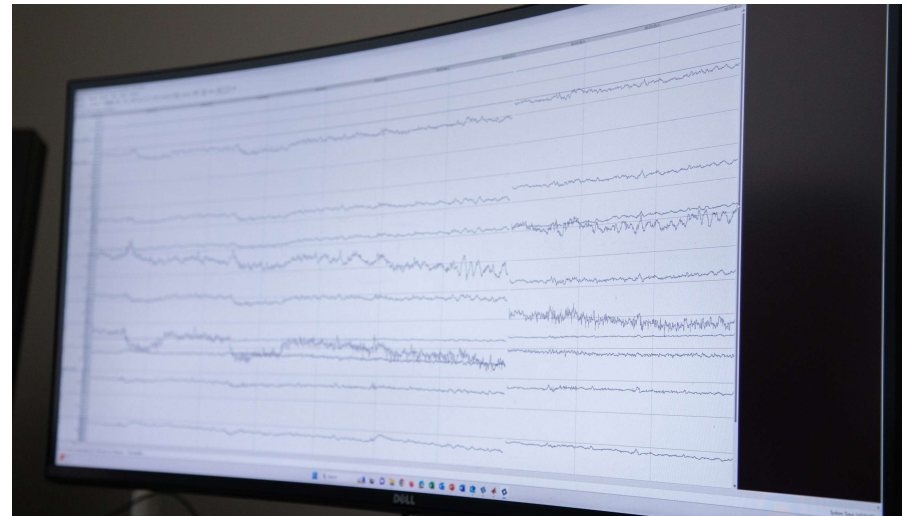
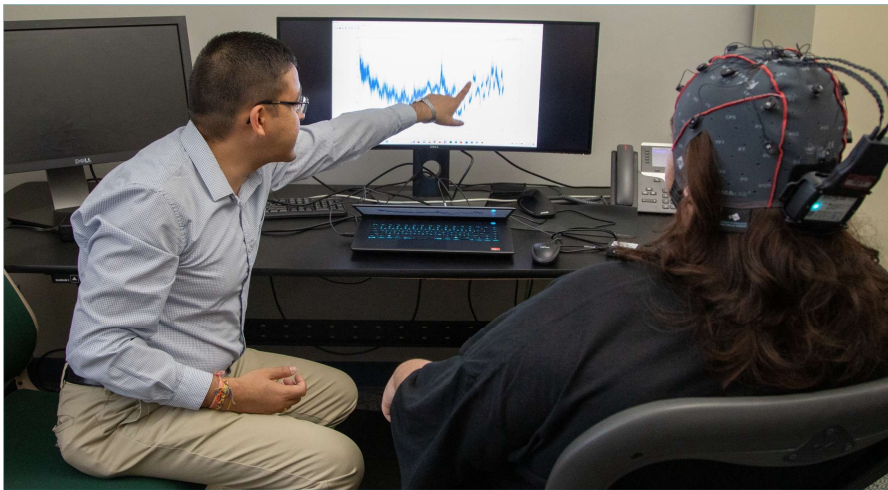


Abdel Deen, Omar & Shieh, Jiann-Shing. (2025). A Multimodal Deep Learning Approach to Intraoperative Nociception Monitoring: Integrating Electroencephalogram, Photoplethysmography, and Electrocardiogram. *Sensors*. 25. 1150. 10.3390/s25041150.

Methods (EEG Data Collection)

- Data Collection Overview :
 - Conducted at the University of Wisconsin-La Crosse BCI Lab
 - Equipment: **g.tec EEG headset** (16-channel cap)
 - Protocols: Eyes Open (EO)
 - Each session: 3 minutes total.
 - Participants: 25 subjects, EO , 3-minute sessions.
 - Two sessions per participant, spaced days to a week apart

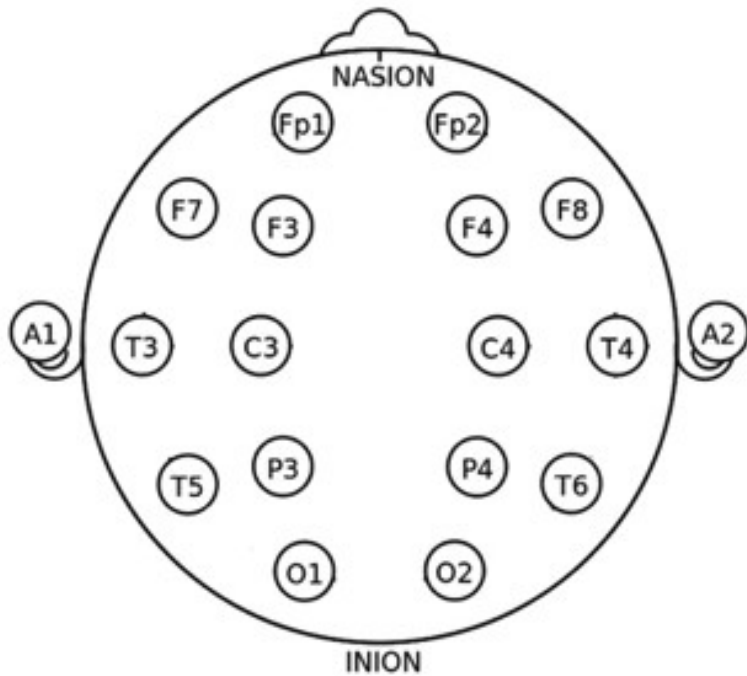
Acquiring Data



**Stimuli For
Acquiring
Data**

Starting in 5 seconds



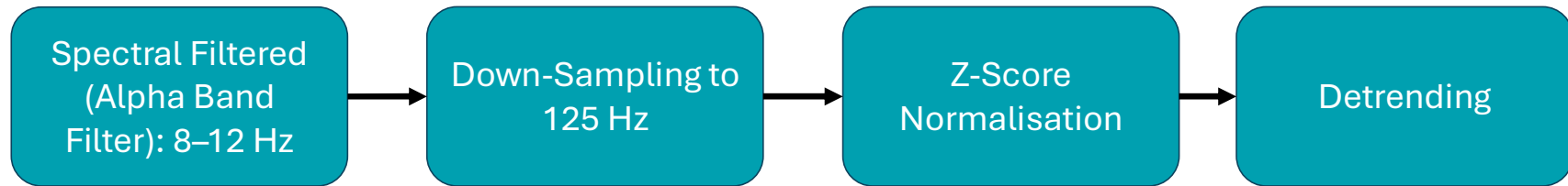


Methods (EEG Data Collection)

- Electrode Placement:
 - Selected nodes: **All nodes except for Fp1 and Fp2 .**
 - Focused on **central** and **occipital** regions for distinct EEG patterns.
 - These areas are known to reflect **cognitive processing** and **visual activity** under open-eye conditions

Öztürk, Hakan & Türe, Mevlüt & Kiylioglu, Nefati & Omurlu, Imran. (2017). The comparison of different dimension reduction and classification methods in EEG signals. Meandros Medical and Dental Journal. 19. 10.4274/meandros.96168.

Methods (Data Pre-Processing)



Methods (Deep Learning Framework)

- Neural networks are computer models inspired by how the brain works. They learn patterns from data by connecting simple units called “neurons.”
- CNNs, or convolutional neural networks, are a type of neural network that’s especially good at processing signals over time—like EEG data.
- Our CNN looks at both how brain signals change over time and how different channels relate to each other.
- It takes in a 14 by 625 segment—14 EEG channels recorded over 5 seconds—and uses layers of processing to figure out which person the signal came from.

Methods (Deep Learning Framework)

Core CNN Layers

Convolutional Layers: Detect patterns in EEG signals, both over time (temporal) and across electrodes (spatial).

ReLU Activation: Adds non-linearity so the network can learn more complex brainwave features.

Pooling Layers: Downsample data to reduce noise and improve generalisation.

Regularisation and Classification

Dropout Layers: Randomly deactivate neurons during training to prevent overfitting.

Fully Connected Layers: Combine features and make the final decision.

Softmax + Classification: Outputs a probability for each subject, choosing the most likely one.

Our Custom-Built CNN for EEG Signals

Input

→ $[14 \times 625 \times 1]$ EEG segment (14 channels, 5s @125Hz, z-scored)

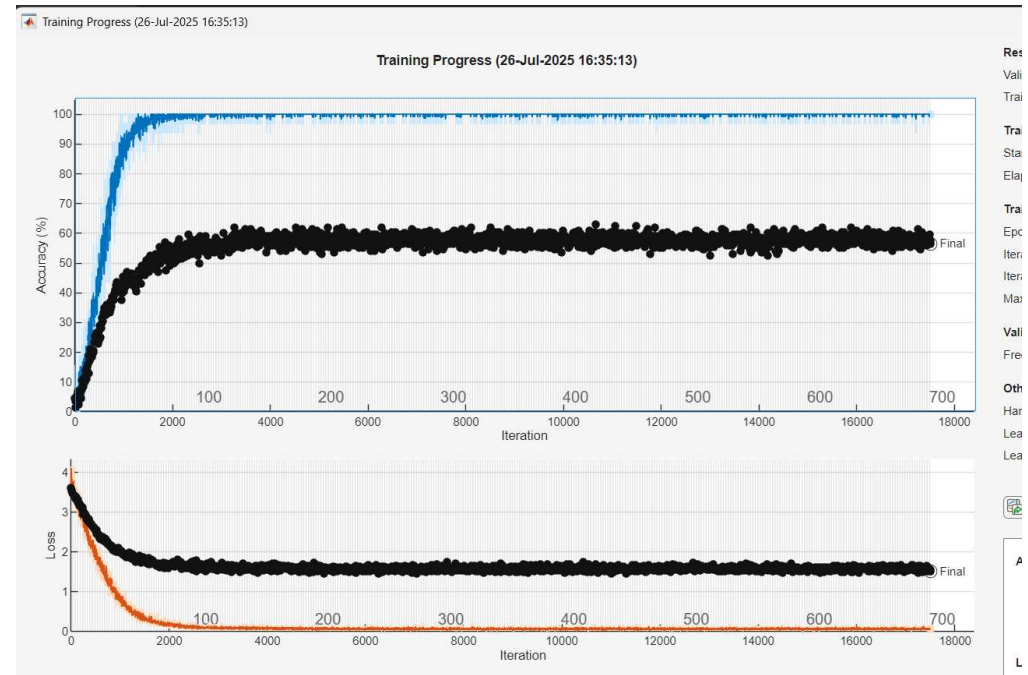
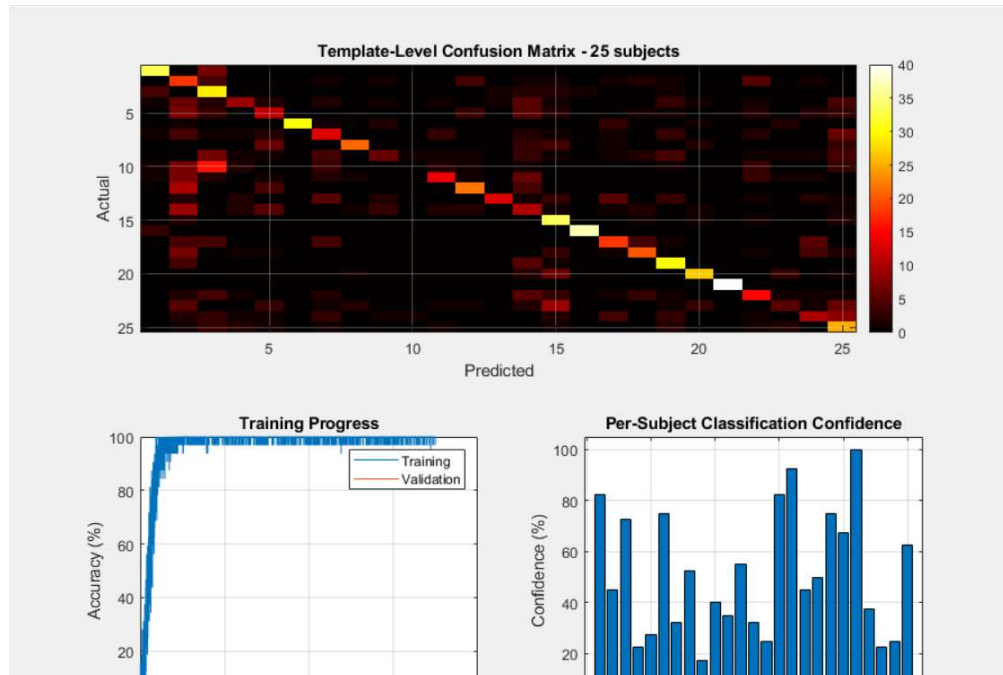
Feature Extraction Stack:

- Conv1 $[5 \times 15]$, BN, ReLU → $[10 \times 611 \times 32]$
- Pool1 $[2 \times 2]$ → $[5 \times 305 \times 32]$
- Conv2 $[5 \times 15]$, BN, ReLU → $[1 \times 291 \times 64]$
- Pool2 $[1 \times 2]$ → $[1 \times 145 \times 64]$
- Conv3 $[1 \times 15]$, BN, ReLU → $[1 \times 131 \times 128]$
- Dropout (30%)
- Pool3 $[1 \times 5]$ → $[1 \times 26 \times 128]$
- Conv4 $[1 \times 13]$, BN, ReLU → $[1 \times 14 \times 256]$
- Conv5 $[1 \times 14]$, BN, ReLU → $[1 \times 1 \times 512]$

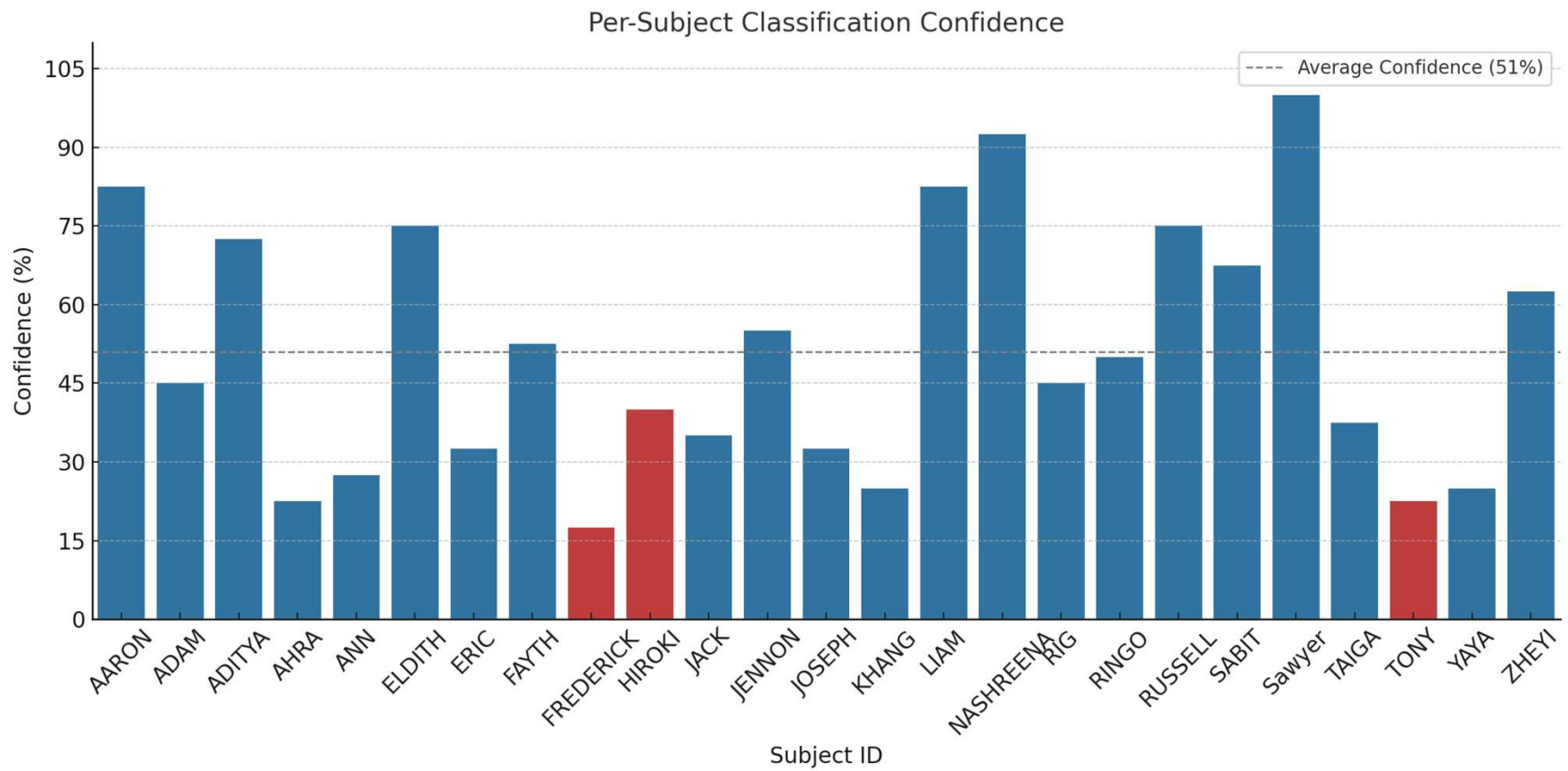
Classifier Head:

- Flatten → [512]
- FC1 [256], BN, ReLU, Dropout (50%)
- FC2 [128], BN, ReLU, Dropout (40%)
- FC Output [25 Subjects], Softmax → Classification

Results and Performance



Results and Performance



Interpreting the Results

Key Observations

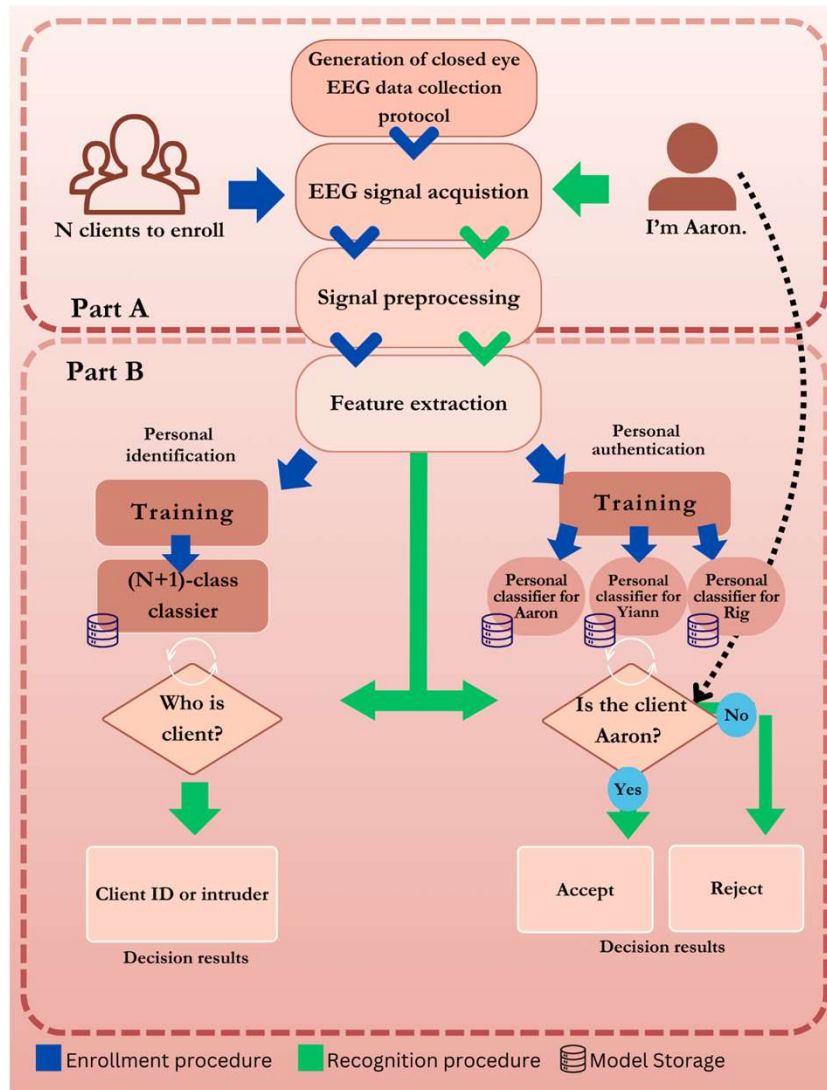
- **Overall Accuracy:**
The model correctly identified 22 out of 25 subjects (88% accuracy) — strong performance for a 25-way classification problem using only 5-second EEG clips.
- **High Confidence Subjects:**
Some subjects (e.g. Sawyer – 100%, Nashreena – 92.5%, Liam – 82.5%) were classified with very high confidence, suggesting consistent EEG patterns across their templates.
- **Misclassifications:**
 - Frederick, Hiroki, and Tony were incorrectly identified.
 - These subjects had lower confidence scores ($\leq 40\%$) — indicating possible overlap in EEG features with others or high intra-subject variability.

Confidence Spread

- Confidence ranged from 22.5% to 100% across subjects.
- Average: 51% – above chance level (4%) in a 25-class problem, showing the model learned meaningful distinctions.
- Lower confidence does not always mean wrong but does signal uncertainty.

What This Tells Us

- EEG segments do carry discriminative features for identity.
- Some users may be inherently easier to identify due to stable EEG patterns.
- More data per subject could help reduce uncertainty and boost confidence.



Brief Overview of Research Project

Limitations and Future Plans

Limitations

- Small dataset: Only 25 subjects.
- Risk of overfitting due to limited variation.
- Some subjects had low confidence (e.g. Ahra, Tony).
- Closed-set only: No impostor (unknown subject) evaluation.

Future Work

- Collect more EEG data across multiple days.
- Add impostor rejection for real-world verification.
- Explore subject-specific fine-tuning.
- Optimise model for lightweight, real-time use.
- Test robustness against spoofing or mimicry attacks.



Conclusion

- Designed a custom **CNN** for biometric authentication using **5-second alpha-band EEG data**.
- Achieved high **subject-level accuracy of 88.0%** through **majority voting** across individual template predictions.
- Model was evaluated **cross-session**, simulating real-world authentication across different recording days.
- Demonstrated the feasibility of **non-invasive EEG-based identification** using deep learning, even with limited training data.
- Results support the use of **temporal-spatial EEG features** as reliable biometric traits.

Any Questions?

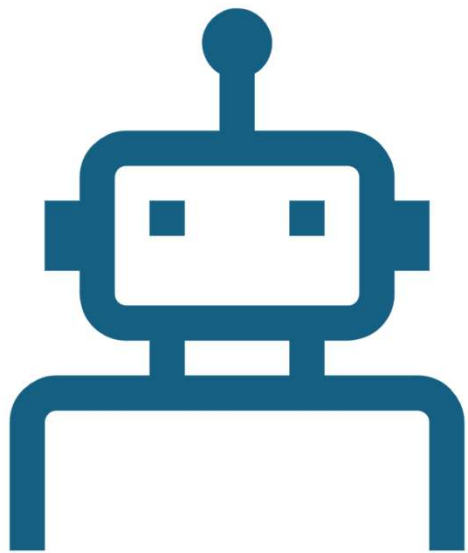
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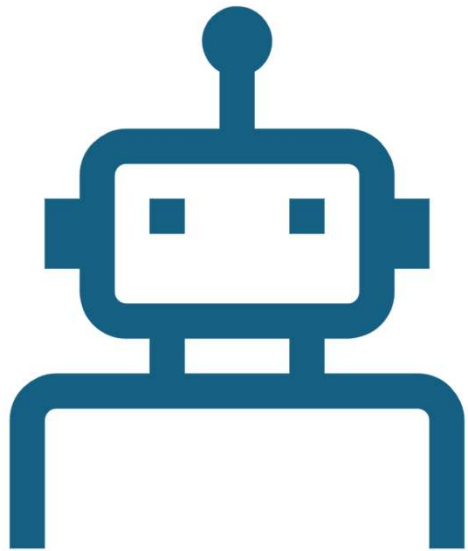
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